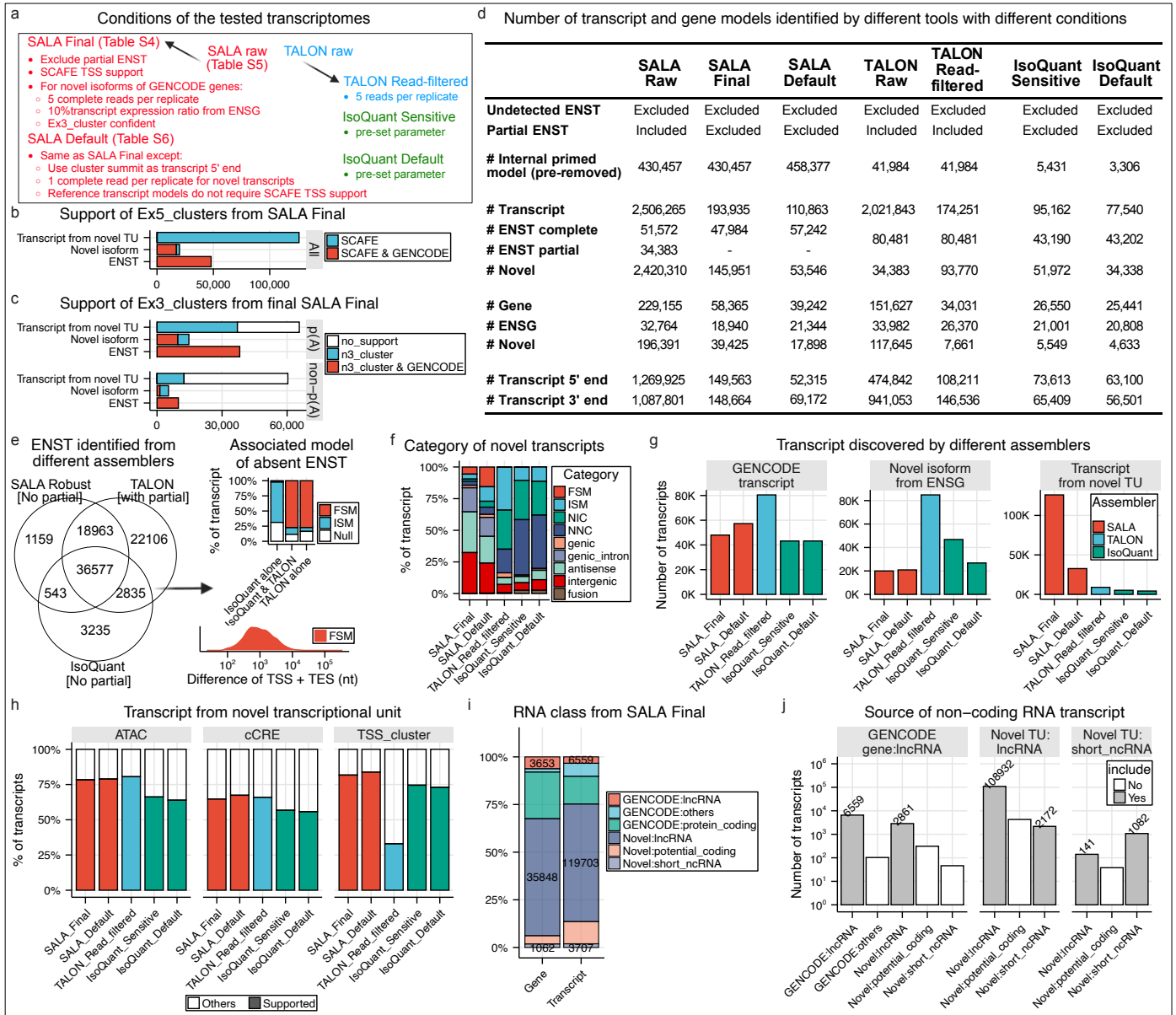




Extended Data Figure 4 | Transcript and gene models identified by SALA.

a, Number of transcript models in the finalized transcriptome with 5' end support from either SCAPE or GENCODE annotations. **b**, Number of transcript models in the finalized transcriptome with 3' end support. Poly(A)-positive (P(A)) transcripts were defined based on a positive poly(A) classifier score or the presence of a polyadenylation signal (PAS) with motif score >3. **c**, Summary of filtering criteria applied to generate different transcriptome versions for benchmarking. **d**, Table comparing the number of transcript and gene models identified by SALA and other transcriptome assembly tools under different filtering conditions. **e**, Venn diagram showing the overlap of detected GENCODE transcript (ENST) across different software tools. Right panel shows the category of transcript model associated to the ENST observed from other tools but not in SALA. Only Full Splice Match (FSM) and Incomplete Splice Match (ISM) were shown. The sum of the distance difference from the two ends between the FSM transcript models and associated ENST was shown in the distribution plot. **f**, Transcript categories assigned to novel transcript models using SQANTI3 across different software outputs. FSM: Full Splice Match, ISM: Incomplete Splice Match, NIC: Novel in Catalog, NNC: Novel Not in Catalog. **g**, Number of transcript models classified into three categories: "GENCODE transcript," "Novel isoform from ENSG," and "Transcript from novel transcriptional unit (TU)." **h**, Proportion of "Transcript from novel TU" models whose 5' ends overlap with open chromatin (ATAC-seq), ENCODE SCREEN candidate *cis*-regulatory elements (cCREs), or SCAPE genuine TSS clusters. **i**, Gene and transcript class assignments in the SALA Final transcriptome. **j**, Number of ncRNA transcript models included in three groups (GENCODE genes, lncRNA from novel TU, and short_ncRNA from novel TU) according to criteria used to define non-coding RNA classes in this study.